S.No	Requirement Category	Requirement Description	Priority (High/ Medium/Low)
1	Authentication	The application provides an open-access dashboard for public analytics. Administrative or data-updating roles must log in securely via standard email and encrypted password validation.	Medium
2	Authorization levels	Guest / Public Analyst: Can view dashboards and simulate data variables. Admin / Data Engineer: Can upload raw HDI.csv source files and overwrite saved model properties.	Medium
3	External interfaces	The system interfaces locally with the Web Browser (frontend) and Python Flask engine (backend), with capabilities to connect to public global development data endpoints via REST APIs.	Low
4	Transactions processing	The application processes immediate input form parameter exchanges, routing user numerical metrics array directly into the active background ML calculation loop.	High
5	Reporting	Generates a clean web dashboard that displays real-time well-being index forecasts. It provides dynamic visual charts indicating how changes in school or health metrics affect final scores.	High
6	Business rules, etc.	Input limits are mathematically bounded based on logical human limits (e.g., Life Expectancy must be between 20 and 100 years). The finalized output score must be clamped strictly between 0.0 and 1.0.	High

7	Compliance to laws or regulations	Follows general data privacy standards; since no sensitive personal identifiable information (PII) of citizens is collected or handled, it adheres strictly to public-domain open data licensing rules.	Low
8	*Other*	Includes an automated exception handler to process empty inputs or trailing spaces without crashing the local application backend engine.	Medium

Solution Requirements

Date	13 JULY 2026
Team ID	xxxxxx
Project Name	Comprehensive measure of well-being
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The local machine learning prediction model logic must calculate and render the wellbeing output index score immediately upon form submission.	Page loads and model predictions calculate in < 1.5 seconds.
2	Scalability	The system must easily process expanding multi-decade historical global data matrices without slowing down system compute pipelines.	Supports uploading datasets up to 100 MB without performance degradation.
3	Security & Data Privacy	Ensure all application source files, background code scripts, and administrative configurations are securely isolated on deployment.	Local system scripts protected via standard file permissions and secure directory structures.
4	Reliability & Availability	The web application dashboard must remain operational and stable during continuous usage or quick analytical parameter re-evaluations.	99% uptime when running on the local or hosted server environment.
5	Usability & Accessibility	The interface must feature a completely clean, responsive, and uncluttered layout that analysts can use without requiring programmatic coding knowledge.	100% accessible via standard web browsers with intuitive numerical inputs.
6	*Other*	Maintainability: The codebase must use clean structural naming practices so future modules or alternative models can be integrated easily.	Modular code splitting (app.py, training scripts) allowing new model updates in < 10 minutes.